# LOGAN YUAN

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## PROFESSIONAL SUMMARY

Software and Biomedical Engineering graduate from McMaster University with hands-on experience building data pipelines, motion analysis systems, and full-stack applications. Proficient in Python, cloud platforms (Azure, AWS), and machine learning frameworks. Passionate about applying engineering principles to real-world problems in healthcare and technology, with a track record of delivering functional prototypes and tools that reduce manual effort.

## EDUCATION

**Bachelor of Software and Biomedical Engineering** Sept 2020 – Jun 2025
*McMaster University, Hamilton, Ontario*

Capstone Project: **Virtual Dragon Boat Coach** Sept 2024 – Apr 2025
*github.com/JamesParkGH/Virtual_Dragon_Boat_Coach*

- Built a motion analysis pipeline in Python that consumes 60 Frames Per Second data from the OpenCap API to evaluate dragon boat athletes' stroke technique and overall form.
- Implemented evaluation algorithms in Python to extract stroke count, joint positions, and joint angles from OpenCap datasets.
- Achieved over 90% accuracy in identifying problematic technique across recorded training sessions.
- Designed a lightweight web interface using Python and HTML to let athletes upload sessions and visualize stroke metrics and feedback, reducing the need for manual video review by coaches.

Course Project: **Neck Pain (NSNP) Monitor** Dec 2022 – Apr 2023

- Prototyped a real-time neck posture monitoring device using an Arduino micro-controller and an orientation sensor to track users' neck angle during everyday office computer use.
- Programmed the micro-controller in C to stream continuous sensor readings via the Arduino serial communication interface to a Python backend at 60 Hz, enabling low-latency monitoring of user posture.
- Designed a Python algorithm that analyzes time-series neck-angle data to identify postures exceeding ergonomic risk thresholds, demonstrating effective performance across all three test users.

## SKILLS

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| **Programming Languages** | Python, Java, JavaScript, C++, HTML |
| **Technical Skills** | Git, Azure, AWS, Linux, PyTorch, Flask, React, PostgreSQL |
| **Certifications** | Microsoft Certified: Azure AI Fundamentals (AI-900) |

## PROJECTS

**Stock Tracker** Dec 2025 – Present
*github.com/yuanq10/StockTracker*

- Created a Python-based stock monitoring tool that delivers automated notifications when configurable conditions are triggered, reducing the time required for daily market analysis.
- Developed a data pipeline that automatically retrieves and processes stock market data using the yfinance API, ensuring the monitoring system stays up to date without requiring users to manually collect market data.
- Designed a user-friendly GUI using CustomTkinter, applying human-computer interaction principles to enable efficient watchlist management, indicator configuration, and data analysis.